Junyi Wu

Ph.D. Candidate · RL Post-Training for Diffusion Models · 2 NeurIPS 2026 Papers
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Education

Purdue University, West Lafayette, IN — Fall 2024 – May 2028 (expected)
Ph.D. in Mathematics. Advisor: Prof. Guang Lin. GPA: 3.97/4.00

University of Chinese Academy of Sciences, Beijing, China — 2017 – 2022
B.S. in Mathematics and Applied Mathematics. GPA: 3.91/4.00. Exchange at National University of Singapore, Spring 2020 (GPA: 4.0/4.0)

Research Experience

Purdue University — West Lafayette, IN
Department of Mathematics | Graduate Research Assistant (Advisor: Guang Lin) — Spring 2025 – Present

– *Diff-Instruct with Diffused Reward: Towards Principled One-step Generator RL* — *NeurIPS 2026*
First author | with Xiaohongshu hi-Lab | Code

- Led DIDR end to end, from theory to multi-GPU training: an **RL post-training** method that aligns **one-step text-to-image generators** with human preferences from **text prompts alone**, no image data.
- Identified **terminal reward domination**, a reward-hacking failure of endpoint-only rewards; fixed it with **Integral-KL RLHF**, proved to share the optimum of standard **KL-regularized RLHF**.
- Derived the **Diffused Reward Score** and a differentiable short-step denoising proxy that carries reward gradients to every noise level.
- Built the **FSDP/BF16** training pipeline on **8 H100s** (48–96 h runs) and scaled it from 2.6B **SDXL** to the 6B **Z-Image** diffusion transformer.
- On the 6B **Z-Image DiT**, lifted one-step **ImageReward 0.39** $\rightarrow$ **1.08** and cut **FID 36.7** $\rightarrow$ **22.1**, beating its **50-step** base on all preference metrics with **50**$\times$ **fewer sampling steps**.
- On SDXL, raised **ImageReward 1.01** $\rightarrow$ **1.10** and **PickScore 23.1** $\rightarrow$ **23.9** over the strongest one-step baseline, beating FLUX-dev and other multi-step models on four preference metrics.

– *COCOA: Action-Conditioned Correction for Vision-Language-Action Policies* — *Under review, ICLR 2027*

- Co-developed a **correction policy** that revises a frozen VLA's action chunks before execution, raising success on three robot policies: Xiaomi-Robotics-0/SimplerEnv **85.5%** $\rightarrow$ **90.4%**, FLOWER/CALVIN **77.8%** $\rightarrow$ **81.9%**, $\pi_{0.5}$ on a real Franka arm **71.7%** $\rightarrow$ **81.7%**.
- Designed the two-stage training: **supervised flow-matching** initialization, then **SRPO preference optimization** on automatically constructed action pairs.

– *Quantum Safe Stochastic Linear Bandits* — *NeurIPS 2026*

- Proved that QRS-COLTS, a quantum Thompson-sampling algorithm for bandits with unknown safety constraints, picks safe actions with high probability and achieves **polylogarithmic regret**.

– *Physics-Informed Deep Operator Kolmogorov–Arnold Networks with Chunk Rational Structure*
First author | Under review, J. Comput. Phys. | Code

- Built PO-CKAN, a **neural PDE solver** in **JAX** (DeepONet + rational KANs + physics losses); cut L2 error by **48%** on Burgers, Eikonal, diffusion-reaction, and fractional PDEs.

– *Patch LegONet: Reusing Low-Frequency Operator Blocks for Localized High-Frequency PDE Dynamics*
Equal contribution | arXiv 2026

- Co-led the method: reused frozen LegONet blocks globally and locally to resolve shocks and thin interfaces; at equal cost, **10–50**$\times$ **more accurate** than DoF-matched global refinement on heat, Allen–Cahn, and Burgers equations.

Projects

Joint Decoding for Masked Discrete Diffusion — Sep 2026
Independent research | Python, PyTorch

- Replaced MDLM's independent per-token decoding with inner **Gaussian flow steps** in embedding space, so tokens unmasked in the same step are sampled jointly.
- On unconditional Sudoku generation (exact validity check), raised the valid-grid rate **6.6%** $\rightarrow$ **22.6%** at matched compute (82 vs. 81 NFE) and **11.5%** $\rightarrow$ **58.4%** at 800 steps.

Self-Speculative Decoding Engine for Looped LLMs — Oct 2026
Independent project | PyTorch, FlashAttention, CUDA Graphs

- Wrote a custom inference engine for **Ouro-1.4B** (4 loops $\times$ 24 shared layers) with a **paged KV cache**, **torch.compile**, and **CUDA graphs**.
- Drafted tokens with the first loop and verified them with the rest: **19.3** $\rightarrow$ **42.4 tok/s (2.19**$\times$**)** on one A100, matching greedy AR output on 254/256 tokens; **1.90**$\times$ with exact match on Recurrent-Llama-3.2.

Skills

Generative Modeling: diffusion/flow models, distillation, one-step generation, text-to-image, DiT, masked discrete diffusion
Post-Training: RLHF, KL regularization, DPO/SRPO, SFT, reward modeling (PickScore, ImageReward, HPSv2.1)
Engineering: Python, C++, PyTorch, JAX, Hugging Face Diffusers/Transformers, timm, NumPy
Tools: PyTorch DDP/FSDP, mixed precision (BF16), FlashAttention, CUDA graphs, NCCL, Git, Linux, Docker, W&B, Claude Code
Service: Reviewer, NeurIPS 2026 | Co-Mentor, Purdue SURF, Summer 2026